



First Semester 2026-27

Part - II

Date: Aug 01, 2026

In addition to Part-I (General Handout for all courses appended to the time table), this portion gives further specific details regarding the course.

Course No: **EEE F211/ECE F211/INSTR F211**

Course Title: **Electrical Machines**

Instructor-in-charge: **HARI OM BANSAL (e-mail: hbansal@pilani.bits-pilani.ac.in)**

Instructors:

Lecture: Hari Om Bansal

Tutorial: Hari Om Bansal, Aditya R Gautam, Hitesh Datt Mathur, Dheerendra Singh, Manmahendra Singh Daksh, Ashish Patel

Practical: Ankiet Saxena, Shubham Tiwary, Surabhi Singh, Shivangi Kaushik, Aakarshan Pratik, Shubham Priyadarshi, Ujjwala Rai, Vikrant Singh

1. Course Description:

Transformer: Constructional features, equivalent circuit and phasor diagram - regulation and efficiency, parallel operation. Three phase transformer connections; Harmonic in transformers; Testing; Phase conversion; Autotransformer. D.C Machines: Construction, armature windings, armature voltage and torque equations, classification. D.C generators, performance characteristics; D.C motors - torque/speed characteristics, speed control and braking. Testing and efficiency. Induction machines: Constructional features and rotating magnetic field. Circuit model and phasor diagram. Steady state characteristics. Testing, starting and speed control. Time harmonics and space harmonics. Wound rotor induction motors, Single phase induction motors - classification and equivalent circuit. Synchronous machines: Constructional features; synchronous generators and motors; equivalent circuit and phasor diagram; power and torque characteristics and capability curves. Parallel operation. Salient pole synchronous machine - phasor diagram and determination of synchronous reactances; starting and speed control of synchronous motors. Special machines- universal motors, Induction generators.

2. Scope and Objective of the Course:

Electrical Machines are the work horse in almost all industries ranging from few watts' power to several mega watt power applications. The scope of this course is to study of working principle & operating characteristics of Electrical Machines for the mode of Generation, and Utilization of Electrical Power. The specific objective of this course is to gain better understanding of various conventional electrical-magnetic devices such as Transformer, DC Machines and AC Machines. The course covers theory as well as detailed experimental component for better understanding of the concepts.

3. Course learning outcomes: The students will learn about the operating principles, constructional features, equivalent circuit diagram, efficiency, testing & control and different type of performance characteristics of transformers, DC machines, induction and synchronous machines. The further details are given in the course plan.

4. Text Books and References:

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|-----|---|
| TB1 | I. J. Nagrath and D. P. Kothari, "Electric Machines", McGraw Hill, 4 th /5 th Edition, 2010/2018. |
| TB2 | Electrical Machines Laboratory Manual (For lab operations) |
| RB1 | Bhag S. Guru and Huseyin R. Hiziroglu, "Electric Machinery and Transformers", Oxford. |
| RB2 | P.S. Bimbhra, "Electrical Machinery", Khanna Publishers. |





5. **Method of conduct of the course:** Offline including Blackboard +ppt

6. **Course Plan:**

Lecture No.	Module/Topics	Learning Outcomes	Reference (TB1)
1-2	Introduction to Poly-phase system	Review of Single /Three-phase system, Concepts of various power & their measurement.	Appendix-I & II
3 -6	Introduction to Magnetic circuits	Review of elementary concepts of Magnetic circuit, their electrical analogy, AC-operation of magnetic circuit, energy stored & losses.	2.1-2.6
7-9	Transformer: Single-Phase	Construction, Principle of Operation, Ideal Transformer, Equivalent Circuit. Practical Transformer, Transformer on No-load and on Load, Equivalent Circuit, Transformer Losses, Testing, Per unit system,	3.1-3.8
10-12	Transformer: Single-Phase	Efficiency, Voltage Regulation, Excitation phenomenon, Harmonics in Transformer	3.9, 3.10
13	Autotransformer	Single Phase Autotransformer	3.11
14	Transformer: Three-Phase	Different 3-phase transformer connections, Zig-Zag transformers	3.13
15-16	Transformer	Parallel Operation of Transformers, Phase Conversion (Scott Connection), Tap changing transformers	3.14, 3.16, 3.17
17-20	Basic Concepts of Rotating Machine	Generated EMF, Rotating Mag. Field-Concepts,	5.1-5.7 & 5.10
21-26	Three-Phase Induction Motors	Constructional features, Principle of operation, Rotating Magnetic Field, Circuit Model, Phasor diagram, Induction Generator, starting and speed control	9.1 - 9.6, 9.8-9.10
27-28	Fractional Kilowatt Motors(Single-phase Induction Motor)	Principle of development of rotating two fields. Types of single phase Induction Motor,	10.1-10.3
29-33	Synchronous Machine	Constructional details, Circuit Model, Operating Characteristics, V –curves, Efficiency	8.1,8.4, 8.9-8.11
34-38	DC Machine	Constructional details, EMF & Torque relationship, Armature reaction , Compensating winding	7.1-7.11
		Speed Control, Speed Regulation, Efficiency of DC Motors, Swinburne's Test and Hopkinson's Test on DC Motors	7.15-7.17, 7.19
39-40	Special Machines	Basics of Brush less DC machine and permanent magnet synchronous motor with their applications	Class notes





7. Evaluation Scheme:

Evaluation Component	Duration	Marks (300)	Date & Time	Nature of component
Class Participation: <i>This component evaluates student engagement during lecture hours and also linked to the attendance in the lecture class</i>	-	20	During lecture hours	OB
Quizzes	15 Min	40	During Tutorial Hours	CB (Best 5 out of 6)
Midsem Test	90 Min	75	10/10 AN1	OB (TB1 and class notes allowed)
Comprehensive Examination	3 Hr	105	16/12 FN	CB
Practical: Attendance, Lab Report & Viva	----	40	Regular Lab Sessions	OB (Except Viva)
Lab Exam (Experiments to be performed in the lab during exam)	1 Hr	20	To be Announced	CB

8. List of Experiments: (Text Book –TB2)

- 1) No load tests on a single- phase transformer(OC/SC test)
- 2) Three phase connection of single-phase transformers
- 3) Speed control of a DC shunt motor
- 4) a. Test on single phase induction motor
b. Open lab experiments
- 5) Synchronization of alternator with infinite bus bar
- 6) Parallel operation of two single phase transformers
- 7) Load test on a DC shunt generator
- 8) Open circuit test on a DC shunt generator
- 9) No load tests on a synchronous machine
- 10) Load tests on three phase induction motor

9. Chamber Consultation Hours: To be announced in the class

10. Notices: All notices will be displayed on Nalanda only

11. Make-Up Policy: No makeup will be granted for Class Participation and Quizzes. Makeup for other components will be granted to genuine cases only.

12. Minimum performance requirement for valid grade: A student should obtain 25% of the average marks of the Class, or 10% of the Course Total Marks, whichever is lower to clear the course. If any student gets the marks lower than the prescribed standard mentioned above, he/she may be awarded NC Report.

Instructor in Charge
Electrical Machines (ECE/EEE/INSTR F211)

